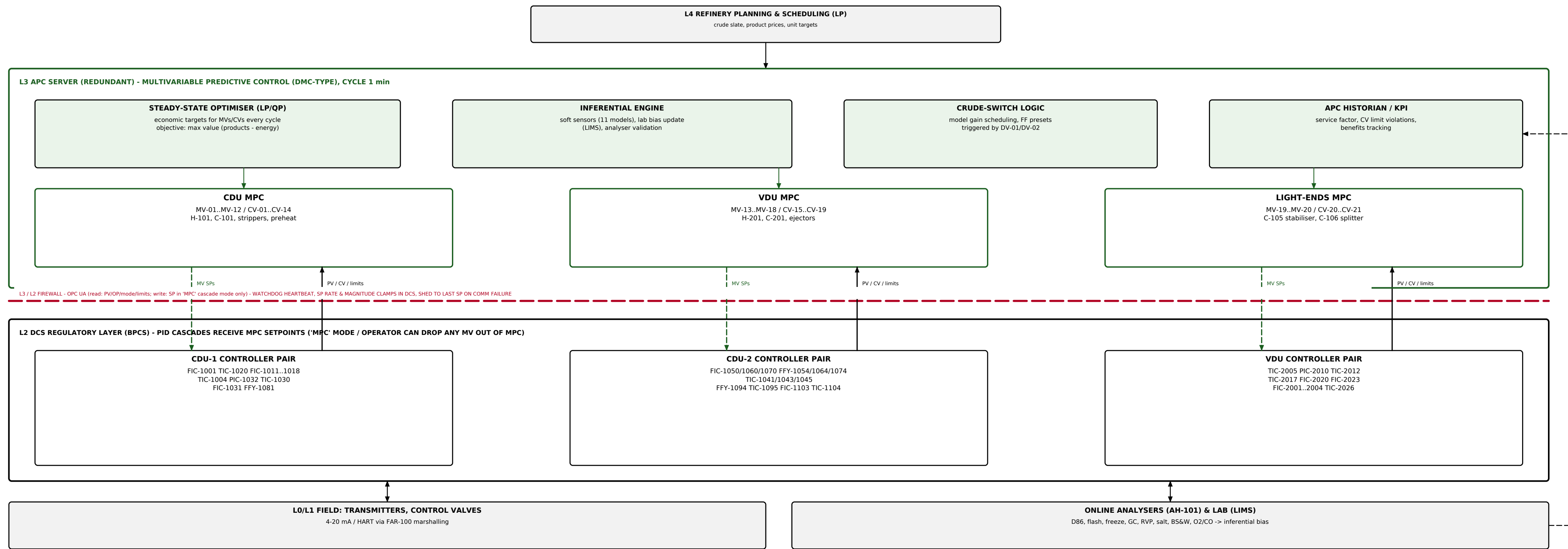




APC DESIGN BASIS

1. Three DMC type MPC applications to redundant L3 APC server (CDU, VDU, light ends) with a common steady-state LP optimiser; 1 minute execution, prediction horizon = 2 – longest settling time at CDU ~ 180 min).
2. Models identified by step testing (3–2 weeks per application) after regulatory tuning audit; crude grade plant scheduling (light / heavy blends), refinery feedstocks (vacuum gas oil, vacuum residue) and product specifications (gasoline, diesel, jet kerosene, naphtha, fuel oils). Economic optimisation (maximise distillate / HVGCO recovery, minimise energy, minimise water, wastewater volume saturation). (3) Process control (distillation column, reactor, heat exchanger).
3. Inferentials run in the APC server (pressure-temperature-composition temperatures, CPD), validated by analysts and lab with slow bias update; ANALYT output > 90%.
4. DCS keeps full authority; MPC writes SPS only to loops in MPC mode; with SPC limits and rates limit; on watchdog failure all loops shed to ANALYT after 60 sec Spd an alarm is raised.
5. Expected benefits (Industrial typical): 3.7 % throughput, 2.5 % energy, 0.5-1 vol % higher-value products vs. current operation.
6. MCV/VDU feeds are FEED-levels; final lost feed in the APC functional specification after the pre-test.

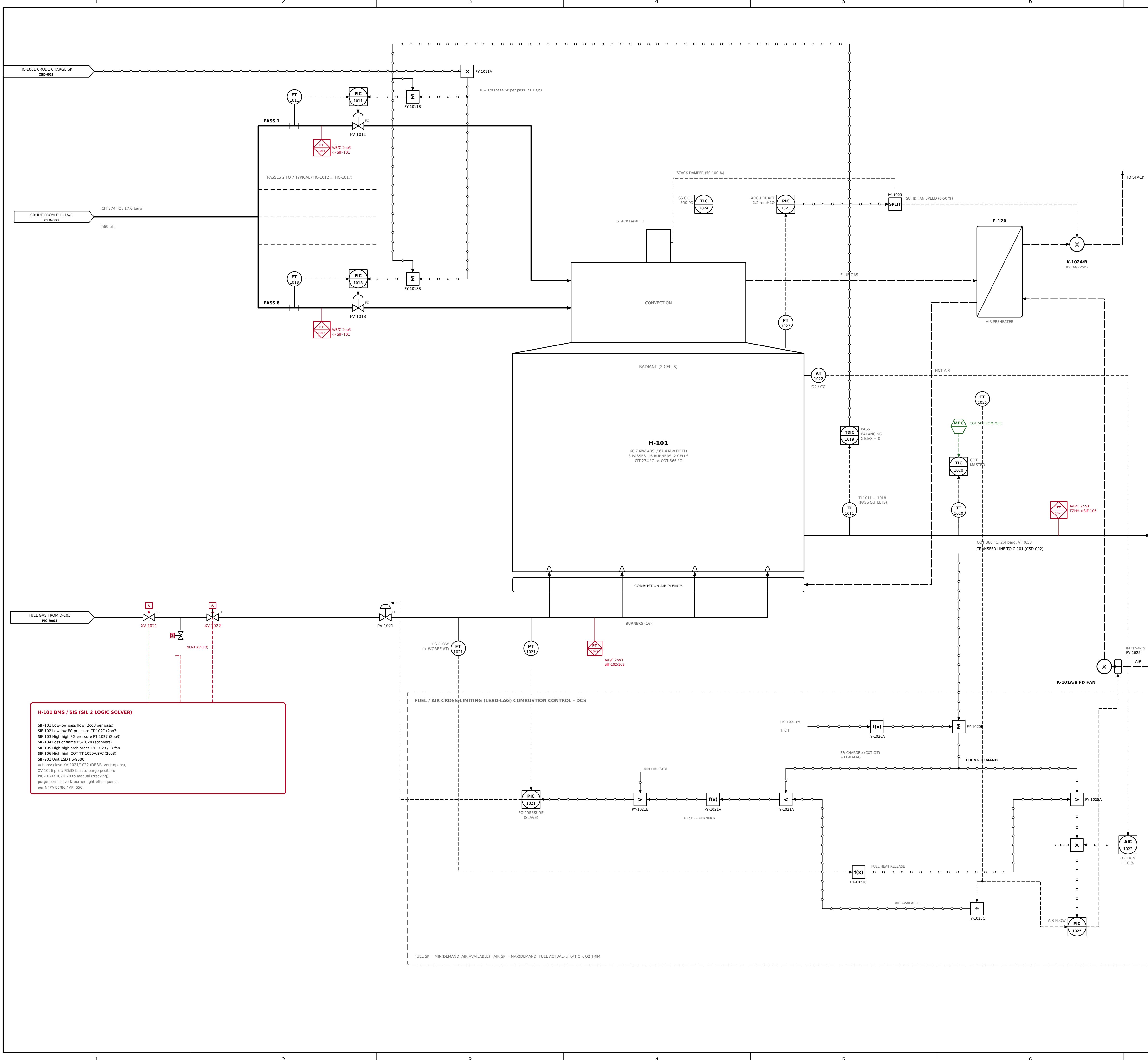
MV	DCS TAG (SP)	DESCRIPTION	RANGE	MPC
MV-01	FIC-1001	Crude charge	331-729 m3/h	CDU
MV-02	TIC-1020	H-101 COT	356-372 °C	CDU
MV-03	TIC-1030	C-101 top temperature (naphtha EP)	124-144 °C	CDU
MV-04	FIC-1050	Kerosene draw	±15 % of design	CDU
MV-05	FIC-1060	Diesel draw	±15 % of design	CDU
MV-06	FIC-1070	AGO draw	±20 % of design	CDU
MV-07	TIC-1041	TPA duty (return T)	73-108 °C	CDU
MV-08	TIC-1043	MPA duty (return T)	150-185 °C	CDU
MV-09	TIC-1045	BPA duty (return T)	214-249 °C	CDU
MV-10	FFY-1081	C-101 stripping steam ratio	6-12 lb/bbl AR	CDU
MV-11	FFY-1054/1064/1074	Side-stripper steam ratios	2-8 lb/bbl	CDU
MV-12	PIC-1032	C-101 OH drum pressure	0.4-1.0 barg	CDU
MV-13	TIC-2005	H-201 COT	386-402 °C	VDU
MV-14	PIC-2010	C-201 top pressure	15-40 mbar(a)	VDU
MV-15	TIC-2012	C-201 top temperature	55-90 °C	VDU
MV-16	TIC-2017	HVGO PA duty (return T)	±20 °C	VDU
MV-17	FIC-2020	Wash oil (>= min wetting)	min-design x 1.3	VDU
MV-18	FIC-2023	C-201 stripping steam	0-1.5 x design	VDU
MV-19	FFY-1094 / TIC-1095	C-105 L/JF, sensitive tray T	±10 % / ±8 °C	LE
MV-20	FIC-1103 / TIC-1104	C-106 reflux, sensitive tray T	±15 % / ±8 °C	LE

CONTROLLED / CONSTRAINT VARIABLES (CV)				
CV	CONTROLLED VARIABLE	MEASUREMENT	TYPE	MPC
CV-01	Naphtha D86 EP	AT-1038 + inferential	range	CDU
CV-02	Kerosene flash point	AT-1055 + inferential	min limit	CDU
CV-03	Kerosene freeze point	inferential (lab bias)	max limit	CDU
CV-04	Diesel D86 T95	AT-1065 + inferential	max / target	CDU
CV-05	Diesel cloud point / flash	inferential (lab bias)	range	CDU
CV-06	AGO D86 T95 / colour	AT-1075 + inferential	max	CDU
CV-07	Kero-diesel / diesel-AGO gap	inferential (5-95 gap)	min	CDU
CV-08	Overflash	FI-1080 / charge	≥= 3 vol %	CDU
CV-09	Section dP (flooding)	PD1-1237	max	CDU
CV-10	H-101 firing / tube metal T	FI-1020B OP, TI skin	max	CDU
CV-11	H-101 arch 02 / draft	AIC-1022, PIC-1023	range	CDU
CV-12	Valve positions	FV-1001/1031, TV bypass, PV-1032A	5-90 %	ALL
CV-13	A-101 duty / OH temperature	TI-A-101 outlet	max	CDU
CV-14	Desalter inlet T	TIC-1004 OP / PV	125-145 °C	CDU
CV-15	LVGO D86 T95 / HVGO CCR, Ni+V	inferential + lab	max	VDU
CV-16	Wash-bed wetting rate	FIC-2020 / bed area	≥= min	VDU
CV-17	C-201 flash zone T / bottoms T	TI flash zone, TIC-2026	max	VDU
CV-18	Ejector load / PV-2010 OP	PIC-2010 OP	max	VDU
CV-19	H-201 firing, col outlet P	PIC-2006 OP, PI	max	VDU
CV-20	LPG C5+ / naphtha RVP	AT-1099 / TIC-1095 infer.	max	LE
CV-21	LN C6+ / HN C5- (IBP)	AT-1108 / AT-1109	max	LE

INFERENTIAL MODELS		
INFERENTIAL (SOFT SENSOR)	MAIN INPUTS	BIAS / VALIDATION
Naphtha D86 EP	TIC-1030, PIC-1032, reflux/draw ratio	AT-1038, lab
Kerosene flash point	Draw T, stripper steam ratio, C-101 P	AT-1055, lab
Kerosene freeze point	Kero draw T, MPA return T, P	lab
Diesel D86 T95	Diesel draw T (PCT), H/C partial pressure	AT-1065, lab
Diesel cloud point	D86 T95 infer., crude API	lab
AGO D86 T95	AGO draw T, overflash, steam ratio	AT-1075, lab
Overflash vol %	Wash-zone liquid F1-1080 / charge	-
LVGO D86 T95	C-201 top P, TIC-2012, LVGO PA	lab
HVGO CCR / Nt+V	Wash rate, flash zone T, HVGO cut	lab
Naphtha RVP (C-105)	TIC-1095, P, feed rate	lab
Atm. residue 370 °C.	Flash zone T/P, steam ratio	lab

DV	DISTURBANCE	SOURCE
DV-01	Crude switch / tank change	Tank-farm signal (OSBL)
DV-02	Crude API / density, salt, BS&W	AT-1048
DV-03	Crude inlet T to heater (CIT)	TI-1226
DV-04	Ambient air temperature	TI-9005 (air coolers)
DV-05	Fuel-gas Wobble / LHV	AT-1021 (FG analyser)
DV-06	MP/LP steam header pressure	PIC-9004
DV-07	AR rate to VDU	FIC-1083 PV (feed-forward)

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
PROPOSED CONTROL SCHEME APC / MPC STRUCTURE (MV / CV / DV)			
DISCIPLINE	STAGE	SCALE	SHEET
INSTRUMENTATION	FEED	NTS	6 OF 6
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-000-IC-CSD-006			A



LEGEND (ISA-5.1 / ISA-5.4)

FIC

0000

DCS (BPCS) function, operator accessible

TC

0000

APC / MPC (L3 computer) function

Σ

Computing block: Σ sum/bias, $\times$ ratio/mult, $<$ low select, $>$ high select, $f(x)$ characteriser

$\rightarrow$

Electrical signal (4-20 mA / HART)

$\rightarrow$

Process line

SIS

0000

SIS / BMS function (SIL rated)

FT

0000

Field-mounted instrument

Control valve (FO/FC = fail open/closed)

Software / data link (DCS internal)

APC setpoint / MPC move

- CONTROL DESCRIPTION
1.

Crude charge master FIC-1001 (CSD-003) sets the base SP of the 8 pass flow controllers FIC-1011..1018 through FY-1011A (K = 1/8, 71.1 t/h per pass).
2.

Pass balancing TDC-1019 compares pass outlet temperatures TI-1011..1018 and computes biases FY-1011B..1018B whose sum is forced to zero, so total charge is unchanged; bias clamped $\pm 10\%$ of pass flow and frozen when any pass FIC is not in CAS.
3.

COT TIC-1020 (SP from MPC) adds to a feed-forward fix of charge x (COT-CTT) with lead lag (FY-1020A/B) to form the firing demand.
4.

Cross-limiting: fuel SP = low select of demand and air available (FY-1021A); air SP = high select of demand and actual fuel heat release (FY-1025A). On load increase air leads fuel; on decrease fuel leads air - no sub-stoichiometric firing.
5.

Fuel heat demand converted to burner pressure SP (burner curve, Wobbe-corrected, PY-1021A); min-fire stop PY-1021B; PIC-1021 manipulates PV-1021.
6.

O2 trim AIC-1022 (arch O2 with CO override) multiplies the air/fuel ratio, limited $\pm 10\%$. FIC-1025 positions the FD fan inlet vanes.
7.

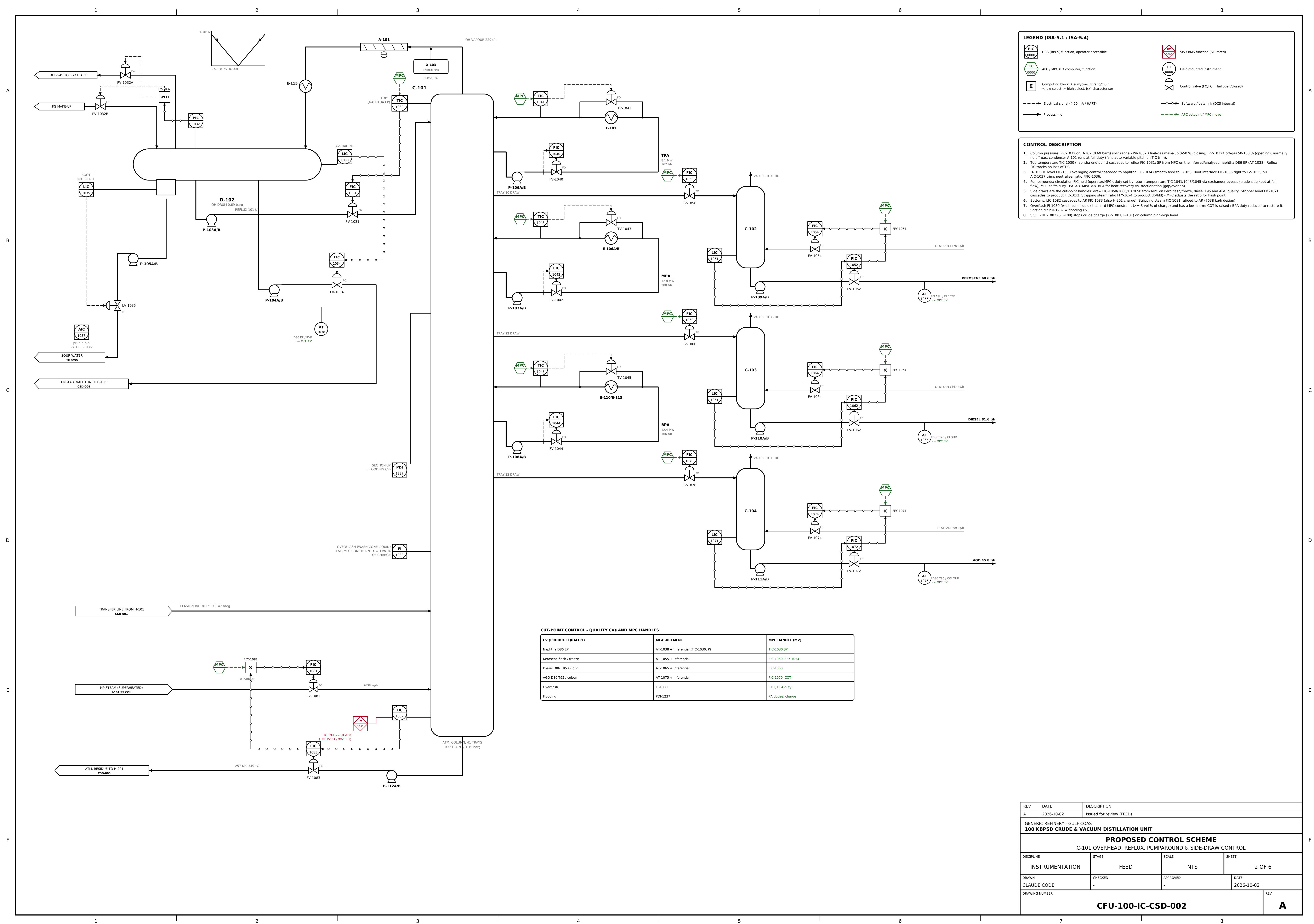
Arch draft PIC-1023 (± 2.5 mmH2O) split range: ID fan speed 0-50 %, stack damper 50-100 % (natural-draft fall-back on ID fan trip).
8.

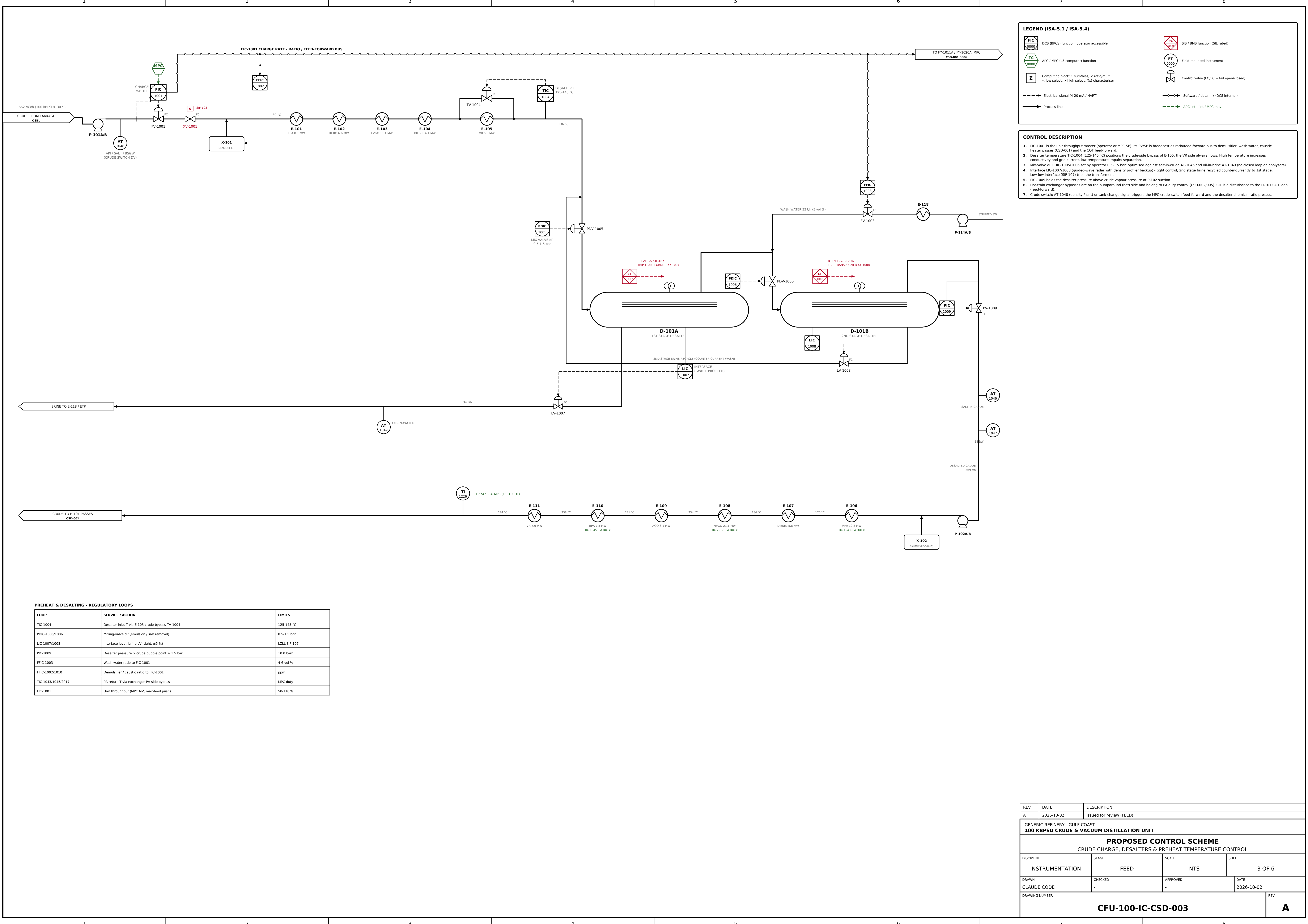
All SIS/BMS functions are independent of the DCS (separate sensors, logic solver and SSOVs).

H-101 BMS / SIS (SIL 2 LOGIC SOLVER)

SIF-101 Low-low pass flow (Zoo3 per pass)
SIF-102 Low-low FG pressure PT-1027 (Zoo3)
SIF-103 High-high FG pressure PT-1027 (Zoo3)
SIF-104 Loss of flame BS-1028 (ccomers)
SIF-105 High-high arch press. PT-1029 / ID fan
SIF-106 High-high COT TT-1020A/B/C (Zoo3)
SIF-901 Unit ESD HS-9000
Actions: close XV-1021/1022 (DB&B, vent opens),
XV-1026 pilot; FD/ID fans to purge position;
PIC-1021/TIC-1020 to manual (tracking);
purge permissive & burner light-off in-sequence
per NFPA 85/86 / API 556.

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
PROPOSED CONTROL SCHEME H-101 COMBUSTION, COT & PASS BALANCING CONTROL			
DISCIPLINE	STAGE	SCALE	SHEET
INSTRUMENTATION	FEED	NTS	1 OF 6
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-100-IC-CSD-001			A





PREHEAT & DESALTING - REGULATORY LOOPS

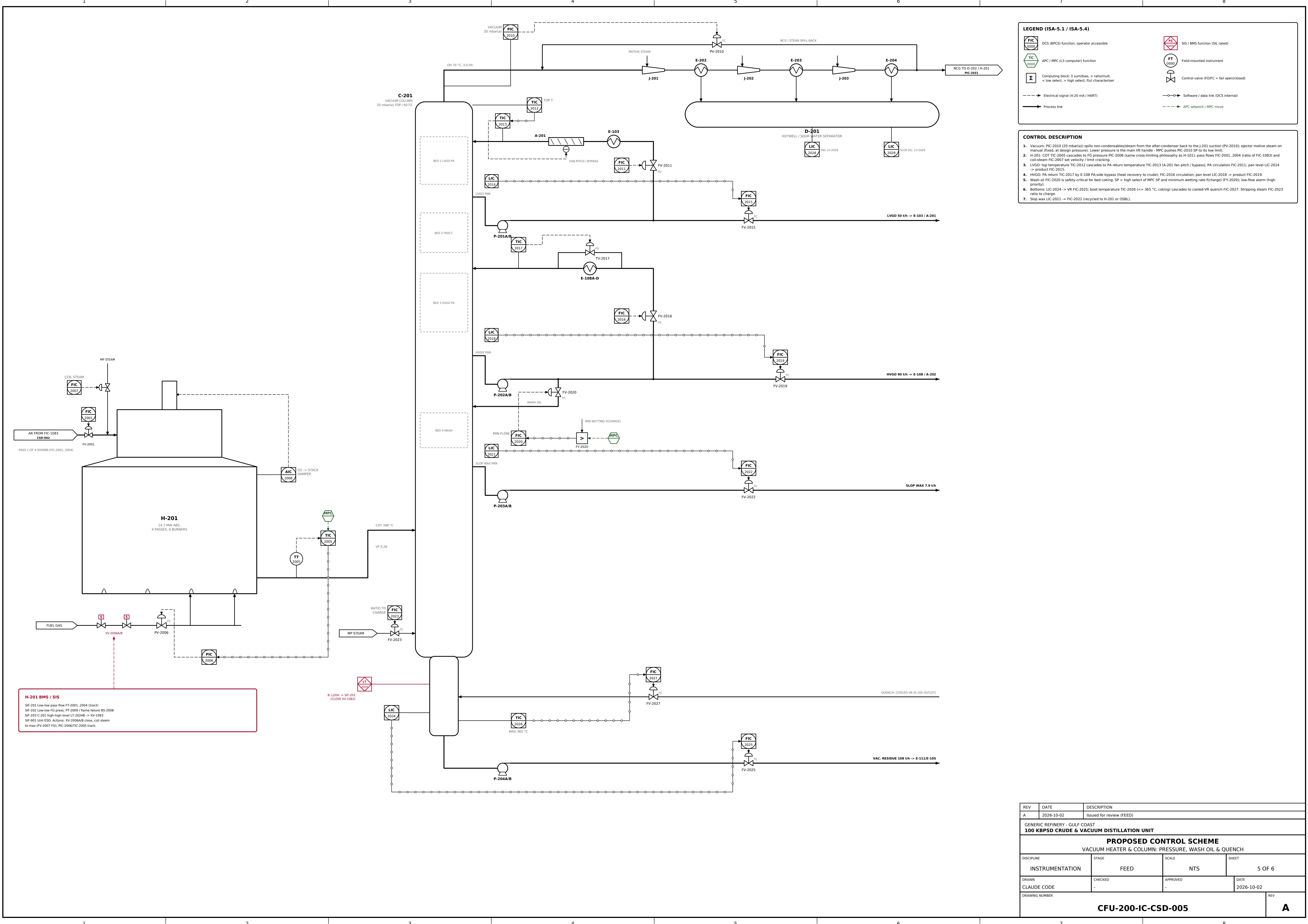
LOOP	SERVICE / ACTION	LIMITS
TIC-1004	Desalter inlet T via E-105 crude bypass TV-1004	125-145 °C
PDIC-1005/1006	Mixing-valve dP (emulsion / salt removal)	0.5-1.5 bar
LIC-1007/1008	Interface level, brine LV (tight, ±5 %)	LZLL SIF-107
PIC-1009	Desalter pressure > crude bubble point + 1.5 bar	10.0 barg
FFIC-1003	Wash water ratio to FIC-1001	4-6 vol %
FFIC-1002/1010	Demulsifier / caustic ratio to FIC-1001	ppm
TIC-1043/1045/2017	PA return T via exchanger PA-side bypass	MPC duty
FIC-1001	Unit throughput (MPC MV, max-feed push)	50-110 %

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
PROPOSED CONTROL SCHEME CRUDE CHARGE, DESALTERS & PREHEAT TEMPERATURE CONTROL			
DISCIPLINE	STAGE	SCALE	SHEET
INSTRUMENTATION	FEED	NTS	3 OF 6
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-100-IC-CSD-003			A

DUAL-COMPOSITION STRATEGY (MPC 2x2 WITH DECOUPLING)

COLUMN	CV (TOP)	CV (BOTTOM)	MV (TOP)	MV (BOTTOM)	PRESSURE
C-105	LPG C5+ (AT-1099)	Naphtha RVP (TIC-1095 / infer.)	FFY-1094 Lf	TIC-1095 SP	PIC-1091 (min P)
C-106	LN C6+ (AT-1108)	HN C5- / IBP (AT-1109)	FIC-1103	TIC-1104 SP	PIC-1100 (float)

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
PROPOSED CONTROL SCHEME STABILISER C-105 & SPLITTER C-106 CONTROL			
DISCIPLINE	STAGE	SCALE	SHEET
INSTRUMENTATION	FEED	NTS	4 OF 6
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-100-IC-CSD-004			A



REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)
GENERIC REFINERY - GULF COAST		
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT		
PROPOSED CONTROL SCHEME		
VACUUM HEATER & COLUMN: PRESSURE, WASH OIL & QUENCH		
DISCIPLINE	STAGE	SCALE
INSTRUMENTATION	FEED	NTS
DRAWN	CHECKED	APPROVED
CLAUDE CODE	-	-
DRAWING NUMBER	CFU-200-IC-CSD-005	
REV	A	